Minia University

Faculty of Engineering

Mechatronics & Industrial Robotics Programm

Electronic Circuits Report
Proteus Simulation Tasks

Course:

Electronic Circuits

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Academic Year

2025 – 2026

Table of Contents

Task 01: Parity Bit Circuit

Muhammed

Objective:

To design and simulate a 4-input parity bit generator using logic gates.

Theory:

A parity bit is used for error detection in digital communication. In even parity, the total number of 1s must be even.

Boolean Expression:

$P = A \oplus B \oplus C \oplus D$

Components Used:

- LOGICSTATE
- XOR Gates / 74LS86
- LOGICPROBE

Truth Table

Task 02: Half Adder Circuit

Muhammed

Objective:

To design and simulate a half adder circuit using logic gates.

Theory:

A half adder is a combinational logic circuit used to add two single-bit binary numbers. It produces two outputs: Sum and Carry.

Boolean Expression:

Sum = $A \oplus B$

Carry = $A \cdot B$

Components Used:

- LOGICSTATE
- XOR Gate / 74LS86
- AND Gate / 74LS08
- LOGICPROBE

Truth Table

Task 03: Full Adder Circuit

Muhammed

Objective:

To design and simulate a full adder circuit using logic gates.

Theory:

A full adder is a combinational logic circuit used to add three binary inputs: A, B, and Cin. It produces two outputs: Sum and Cout.

Boolean Expression:

$Sum = A \oplus B \oplus Cin$

$Cout = A \cdot B + Cin \cdot (A \oplus B)$

Components Used:

- LOGICSTATE

- XOR Gates / 74LS86

- AND Gates / 74LS08

- OR Gate / 74LS32

- LOGICPROBE

Truth Table

Task 04: Subtractor Circuit and Applications

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table

Task 05: 8-Bit Adder Using 74HC283

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 06: 4-Bit Comparator from Scratch

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 07: 2-Bit Comparator Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 08: Decoder Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 09: Encoder Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 10: Multiplexer Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 11: Demultiplexer Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 12: S-R Latch Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 13: R-S Latch Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 14: D Latch Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Task 15: J-K Latch Circuit

Objective:

Theory:

Boolean Expression:

Components Used:

Truth Table:

Appendices

A.1 Parity Bit Circuit Truth Table

Truth Table – 4-Input Even Parity Generator

$$P = A \oplus B \oplus C \oplus D$$

P is the parity bit output for even parity.

A	B	C	D	P
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

The output P is 1 when the number of 1s in A, B, C, and D is odd, so the total number of 1s becomes even.

A.2 Half Adder Circuit Truth Table

Truth Table – Half Adder

$$Sum = A \oplus B$$

$$Carry = A \cdot B$$

A half adder adds two 1-bit binary inputs and produces Sum and Carry outputs.

A	**B**	**Sum**	**Carry**
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

The Sum output is given by XOR, while the Carry output is given by AND.

A.3 Full Adder Circuit Truth Table

Truth Table – Full Adder

$$\text{Sum} = A \oplus B \oplus C_{in}$$

$$\text{Cout} = A \cdot B + C_{in} \cdot (A \oplus B)$$

A full adder adds three 1-bit binary inputs and produces Sum and Carry outputs.

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The Sum output is given by XOR, while the Cout output represents the carry output.